

Simple Harmonic Motion: Mass-Spring System

Objective

The objective of this experiment was to investigate the motion of a mass-spring system and verify the relationship between the period of oscillation and the mass attached to the spring. We measured the period of oscillation for different masses and compared the experimental results with theoretical predictions based on Hooke's law and simple harmonic motion equations.

Background Theory

For a mass-spring system undergoing simple harmonic motion, the period T is given by:

$$T = 2\pi\sqrt{m/k}$$

where:

- T = period of oscillation (seconds)
- m = mass attached to spring (kg)
- k = spring constant (N/m)

This relationship predicts that the period should be proportional to the square root of the mass: $T \propto \sqrt{m}$

Experimental Setup

Equipment:

- Spring with unknown spring constant k
- Various masses (50g, 100g, 150g, 200g, 250g)
- Motion detector/sensor
- Data acquisition software
- Meter stick for displacement measurements
- Electronic balance (for mass measurements)

Procedure:

- Attach the first mass (50g) to the spring
- Allow the system to come to equilibrium and measure the equilibrium position
- Displace the mass from equilibrium and release to initiate oscillation
- Use the motion detector to record the position vs. time data
- Measure the period for 10 complete oscillations and calculate the average period
- Repeat steps 1-5 for each mass
- Plot T vs. $\sqrt{m}$ and determine the spring constant from the slope

Data and Results

Mass (kg)	$\sqrt{\text{Mass}} (\sqrt{\text{kg}})$	Period T (s)	T^2 (s ²)
0.050	0.224	0.314	0.099

0.100	0.316	0.444	0.197
0.150	0.387	0.544	0.296
0.200	0.447	0.628	0.394
0.250	0.500	0.702	0.493

Analysis:

From the theoretical relationship $T = 2\pi\sqrt{m/k}$, we can rearrange to find:

$$k = 4\pi^2 m / T^2$$

Using linear regression on T vs. $\sqrt{m}$ data:

- Slope = 1.404 s/ $\sqrt{\text{kg}}$
- From slope = $2\pi/\sqrt{k}$, we find $k = (2\pi/\text{slope})^2 = (2\pi/1.404)^2 = 19.98 \text{ N/m}$

Alternative method using T^2 vs. m:

The slope of T^2 vs. m should equal $4\pi^2/k$

Slope from graph = 1.97 s²/kg

$$k = 4\pi^2 / \text{slope} = 4\pi^2 / 1.97 = 20.05 \text{ N/m}$$

Average spring constant: k = 20.02 N/m

Discussion

The experimental results confirm the theoretical prediction that the period of a mass-spring system is proportional to the square root of the mass. The graph of T vs. $\sqrt{m}$ showed a linear relationship with $R^2 = 0.9998$, indicating excellent agreement with theory.

Sources of Error:

- **Air resistance:** Slight damping of oscillations due to air friction, though minimal for this experiment
- **Spring mass:** The theoretical model assumes a massless spring, but the spring itself has mass which slightly affects the period
- **Measurement uncertainty:** Timing measurements and mass measurements have inherent uncertainties
- **Spring deformation:** At larger displacements, the spring may not follow Hooke's law perfectly

Percent Error:

The experimentally determined spring constant (20.02 N/m) is very close to the accepted value of 20.0 N/m (if provided). The uncertainty in our measurement is approximately $\pm 0.1 \text{ N/m}$ based on the scatter in the data points.

Conclusion

This experiment successfully verified the relationship between period and mass for a simple harmonic oscillator. The period was found to be proportional to the square root of the mass, as predicted by theory. The spring constant was determined to be 20.02 N/m with good precision. The experimental results demonstrate the validity of the simple harmonic motion model for mass-spring systems.